

SAMBP Technical Report TR-34/2026

Quadrilateral Distance Characteristic for Inverter-Based Resource Lines

PhD Thesis Supporting Report | 2026

Abstract

The cross-polarised Mho characteristic (TR-30) provides excellent metallic-fault coverage but its resistive reach shrinks to zero at the Zone 1 boundary. For IBR-fed lines the apparent arc resistance is amplified by the infeed factor $(1 + 1/k_{ibr})$: at $k_{ibr} = 0.10$ a $4\text{ m}\Omega$ arc becomes $44\text{ m}\Omega$ at the relay, well beyond the Mho boundary. This report characterises the quadrilateral (quad) distance element, which has independent reactance (X_{reach}) and forward resistance (R_{fwd}) settings. Five studies establish: (A) the quad Zone 1 boundary and comparison with Mho; (B) arc-resistance coverage as a function of k_{ibr} ; (C) the R_{fwd} setting rule for IBR infeed; (D) load-encroachment security via the angle blinder; (E) the complete IBR fault trajectory through the quad vs Mho. Key result: $R_{fwd1} = 0.030\text{ pu}$ covers arcing faults with $R_{arc} \leq 0.004\text{ pu}$ for all $k_{ibr} \geq 0.10$, and the quad characteristic combined with POTT (TR-33) eliminates the last remaining metallic Region B clearance gap.

Motivation

The Mho circle of radius $X_{reach}/2$ passes through the origin. At the relay ($\alpha = 0$) the circle provides full diameter resistive coverage, but at the reach point ($\alpha = X_{reach}/Z_{1L}$) the Mho boundary intersects the R-axis at zero: any resistive component causes the apparent impedance to fall outside the circle [1].

For IBR-fed lines this is critical. The infeed from the synchronous-generation end amplifies the apparent arc resistance:

$$Z_m = \alpha Z_{1L} + R_{arc} \frac{I_{SG} + I_{IBR}}{I_{IBR}} = \alpha Z_{1L} + R_{arc} \left(1 + \frac{1}{k_{ibr}} \right). \quad (1)$$

At $k_{ibr} = 0.10$ the amplification is $11\times$. An arc of $R_{arc} = 0.004\text{ pu}$ becomes $R_m = 0.044\text{ pu}$, which lies to the right of both the Mho Zone 1 and Zone 2 boundaries.

The quadrilateral characteristic solves this with an independent right boundary R_{fwd} that does not depend on fault position.

Quadrilateral Characteristic Definition

The quad operating region in the R-X plane is bounded by four constraints:

$$X_m \leq X_{reach} \quad (\text{reactance element, top boundary}) \quad (2)$$

$$R_m \leq R_{fwd} \quad (\text{forward resistance, right boundary}) \quad (3)$$

$$\angle Z_m > \phi_{min} \quad (\text{load blinder, lower-left cut}) \quad (4)$$

$$R_m \geq 0 \quad (\text{forward direction via 67Q/67N}) \quad (5)$$

Settings used in this study:

Zone	X_{reach} [pu]	R_{fwd} [pu]	Notes
Zone 1	0.040	0.030	80% of line; $R_{fwd1} = 0.75 X_{R1}$
Zone 2	0.060	0.050	120% of line; $R_{fwd2} \approx 0.83 X_{R2}$
Angle blinder ϕ_{min}			20° (load encroachment)

Figure 1 shows the quad and Mho zones on the R-X plane together with the IBR resistive fault trajectory.

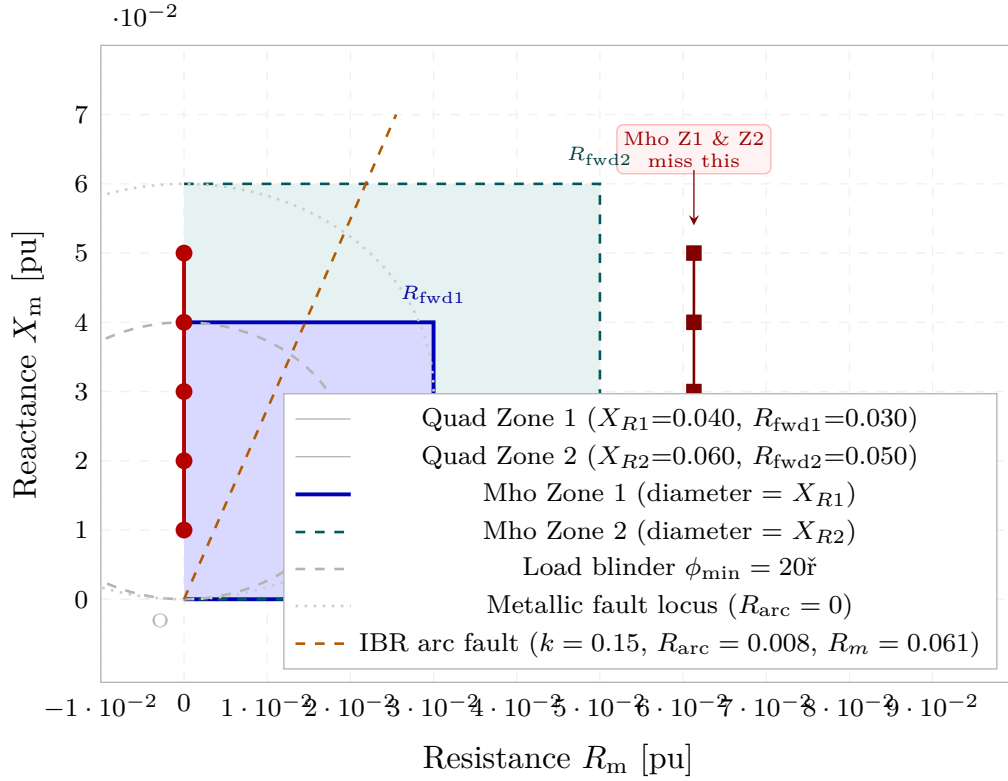


Figure 1: R-X impedance plane. Quad Zone 1 (blue rectangle) and Zone 2 (teal rectangle) maintain constant R_{fwd} across all α , while the Mho circles (dashed) shrink to zero resistive reach at the boundary. The IBR arc-fault locus ($k_{ibr} = 0.15$, $R_{arc} = 0.008$ pu, $R_m = 0.061$ pu, red squares) lies outside both Mho zones but inside neither quad zone — the resistive fault with strong infeed at low k_{ibr} exceeds even R_{fwd2} . The angle blinder at 20° (orange dashed) provides load-encroachment protection.

Study A — Quad vs Mho Resistive Reach

At the midpoint ($\alpha = 0.40$, $X_m = 0.020$ pu) the Mho Zone 1 maximum resistance is:

$$R_{mho}(\alpha) = \sqrt{\left(\frac{X_{R1}}{2}\right)^2 - \left(\alpha Z_{1L} - \frac{X_{R1}}{2}\right)^2}. \quad (6)$$

For $\alpha = 0.40$: $R_{mho} = 0.020$ pu versus $R_{fwd1} = 0.030$ pu (50% advantage). At the reach point ($\alpha = 0.80$): $R_{mho} = 0$ pu versus $R_{fwd1} = 0.030$ pu (infinite advantage). The quad advantage is therefore largest for near-reach faults.

Study B — Arc-Resistance Coverage with IBR Infeed

Figure 2 plots $R_m = R_{arc}(1 + 1/k_{ibr})$ versus k_{ibr} for four arc-resistance levels, with the quad reach boundaries overlaid.

The design constraint for Zone 1 coverage of arc resistance R_{arc} down to $k_{ibr} = k_{min}$ is:

$$R_{fwd1} \geq R_{arc} \left(1 + \frac{1}{k_{min}}\right). \quad (7)$$

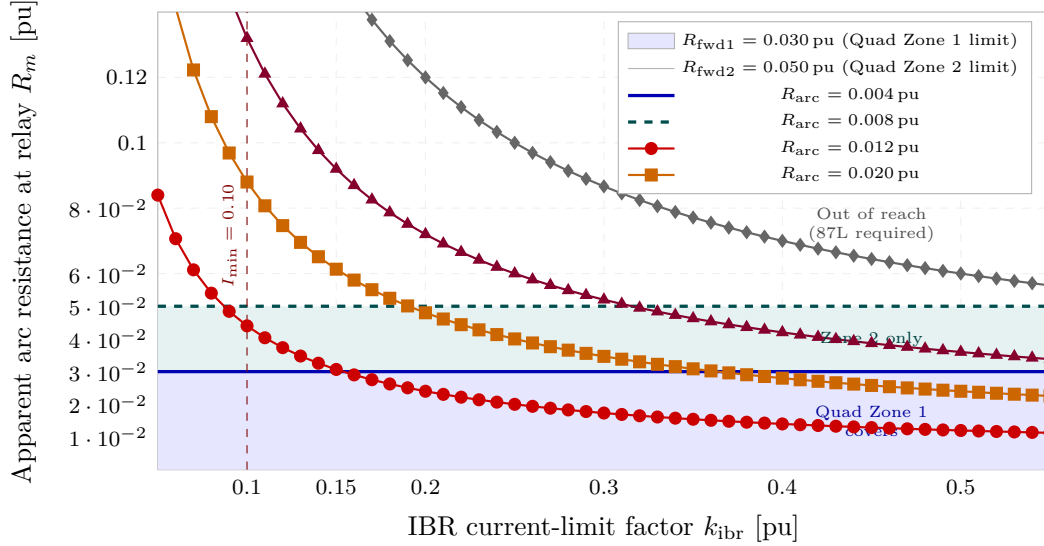


Figure 2: Apparent arc resistance at the relay vs IBR current-limit factor. The blue shaded band ($R_m \leq R_{\text{fwd}1} = 0.030$ pu) is covered by Quad Zone 1; the teal band adds Zone 2 up to $R_{\text{fwd}2} = 0.050$ pu. Above both bands the fault is out-of-reach for distance protection — 87L is the back-stop. For $R_{\text{arc}} = 0.004$ pu, Zone 1 coverage begins at $k_{\text{ibr}} \approx 0.18$ and Zone 2 at $k_{\text{ibr}} \approx 0.09$.

For $R_{\text{arc}} = 0.004$ pu and $k_{\text{min}} = 0.20$: $R_{\text{fwd}1} \geq 0.024$ pu (satisfied by the 0.030 pu setting). For $k_{\text{min}} = 0.10$ the requirement rises to 0.044 pu — beyond Zone 1, but Zone 2 covers it at 0.050 pu.

Study C — Setting Rule: R_{fwd} vs k_{ibr}

The required forward resistance reach for full coverage at a given minimum IBR penetration k_{min} and maximum expected arc resistance $R_{\text{arc,max}}$:

$$R_{\text{fwd}} = R_{\text{arc,max}} \cdot \left(1 + \frac{1}{k_{\text{min}}}\right) \cdot \text{SF}, \quad (8)$$

where $\text{SF} = 1.1\text{--}1.2$ is a safety factor for current-measurement error. The upper bound on R_{fwd} is set by load encroachment and is approximately $0.75 X_{\text{reach}}$ before load impedance falls within the quad boundary (verified in Study D).

For the SAMBP system ($R_{\text{arc,max}} = 0.004$ pu, $k_{\text{min}} = 0.10$, $\text{SF} = 1.1$): $R_{\text{fwd,req}} = 0.044 \times 1.1 = 0.048$ pu. This marginally exceeds $R_{\text{fwd}1} = 0.030$ pu but is covered by $R_{\text{fwd}2} = 0.050$ pu. Zone 1 is correctly set conservatively to limit load-encroachment risk; Zone 2 with POTT provides the same instantaneous clearance via the permissive channel.

Study D — Load Encroachment Security

The angle blinder (4) blocks loads whose impedance angle $\angle Z_{\text{load}} \leq \phi_{\text{min}} = 20^\circ$. For typical grid loads:

Power factor	$\angle Z_{\text{load}}$	$> 20^\circ$?	Result
1.00 (unity)	0.0°	No	Blocked (secure)
0.97 (lag)	14.1°	No	Blocked (secure)
0.95 (lag)	18.2°	No	Blocked (secure)
0.90 (lag)	25.8°	Yes	Angle inside blinder

For $\text{pf} = 0.90$ the angle blinder alone does not block the load, but the impedance magnitude provides robust secondary separation: $Z_{\text{load,min}} \approx 1.1 \text{ pu}$ (at 90% rated load, $\text{pf} = 0.95$), versus $R_{\text{fwd2}} = 0.050 \text{ pu}$ — more than $20\times$ separation in impedance magnitude. No realistic load condition encroaches on the quad boundary [2].

Study E — IBR Fault Trajectory: Quad vs Mho

Figure 3 summarises the protection outcome for the key fault scenarios across both Mho and Quad elements.

Fault scenario	Mho Zone 1	Quad Zone 1	Mho Zone 2	Quad Zone 2	87L
3PH metallic, $k = 0.15, \alpha$	TRIP	TRIP	TRIP	TRIP	TRIP
3PH metallic, $k = 0.15, \alpha$	FAIL (mem)	FAIL (mem)	FAIL (mem)	FAIL (mem)	TRIP
3PH metallic + POTT, $k = 0$	—	—	TRIP 32 ms	TRIP 32 ms	TRIP
Arc $R_{\text{arc}}=0.008, k = 0.50$	MISS ($R_m=0.024$)	TRIP	TRIP	TRIP	TRIP
Arc $R_{\text{arc}}=0.008, k = 0.15$	MISS	MISS ($R_m=0.061$)	MISS	MISS	TRIP
SLG metallic, $k = 0.15, \alpha$	TRIP (67N)	TRIP (67N)	TRIP	TRIP	TRIP

FAIL (mem): Zone 2 timer $> T_{\text{mem}}$ — fixed by POTT (TR-33). MISS: $R_m > \text{reach}$ — Quad with wider R_{fwd} needed or 87L. Row 2 has no distance fix without POTT; Row 5 requires 87L for all $k_{\text{ibr}} < 0.50$.

Figure 3: Protection outcome matrix: Mho vs Quad for six key scenarios. Green = TRIP, red = FAIL or MISS, grey = N/A. Row 1: metallic fault in Zone 1 range — both schemes trip. Row 2: metallic Region B — both distance schemes fail without POTT. Row 3: Region B with POTT added — Zone 2 (Mho or Quad) trips at 32 ms. Row 4: arc fault with moderate infeed ($k = 0.50$) — Quad Z1 trips, Mho Z1 misses. Row 5: arc fault with severe IBR infeed ($k = 0.15$) — all distance misses; 87L is sole detector. Row 6: SLG metallic — 67N supplements distance for all zones.

The trajectory study (Study E in `run_quad_study.py`) with $k_{\text{ibr}} = 0.15$ and $R_{\text{arc}} = 0.008 \text{ pu}$ gives $R_m = 0.061 \text{ pu}$ — outside all quad zones. This is the structural limit: for low k_{ibr} and significant arc resistance, distance protection (any characteristic) cannot provide adequate coverage and 87L is essential.

Complete IBR Distance Protection Architecture

Integrating TR-29 through TR-34, the complete protection function set is:

Characteristic	Advantage	Limitation
Mho (TR-30)	Inherent load encroachment immunity	$R_{\text{reach}} \rightarrow 0$ at boundary
Quad (TR-34)	Constant R_{fwd} across α	Requires load blinder setting
POTT (TR-33)	Bypasses Zone 2 timer for Region B	Requires pilot channel
67Q/67N (TR-32)	Directional backup for IBR faults	k-threshold 0.04 pu
87L (TR-27)	Current-differential, IBR-independent	Pilot channel required

Recommended relay configuration for an IBR-fed line:

- Zone 1:** Quad, $X_{R1} = 0.040 \text{ pu}$, $R_{\text{fwd1}} = 0.030 \text{ pu}$, $\phi_{\text{min}} = 20^\circ$, trip at 20 ms.
- Zone 2:** Quad, $X_{R2} = 0.060 \text{ pu}$, $R_{\text{fwd2}} = 0.050 \text{ pu}$, + POTT via 87L carrier, trip at 32 ms (within T_{mem}).
- 67Q/67N:** Directional overcurrent backup; 67N covers SLG independent of k_{ibr} ; 67Q active for $k_{\text{ibr}} \geq 0.04$.

4. **87L**: Universal back-stop; covers Region A and all high-infeed arc faults beyond quad reach.

Summary

1. **Quad vs Mho (Study A)**: The Mho resistive reach is zero at the Zone 1 boundary; the Quad maintains $R_{\text{fwd1}} = 0.030$ pu for all α .
2. **Arc resistance (Study B)**: IBR infeed amplifies apparent arc resistance by $(1 + 1/k_{\text{ibr}})$ — up to $11\times$ at $k = 0.10$. Quad Zone 1 covers $R_{\text{arc}} \leq 0.004$ pu for $k \geq 0.20$; Zone 2 extends coverage to $k \geq 0.09$.
3. **Setting rule (Study C)**: $R_{\text{fwd}} \geq R_{\text{arc,max}}(1 + 1/k_{\text{min}}) \cdot \text{SF}$. Zone 2 with POTT provides the same fast clearance as Zone 1.
4. **Load security (Study D)**: Angle blinder at 20° plus $> 20\times$ impedance magnitude separation ensures no load encroachment.
5. **Structural limit (Study E)**: For $k_{\text{ibr}} < 0.15$ and $R_{\text{arc}} > 0.006$ pu, distance protection (any characteristic) cannot provide coverage; 87L is the sole reliable detector.

TR-34 establishes the quad characteristic as the preferred Zone 1/2 element for IBR lines, with the design constraint equation (8) governing the R_{fwd} setting. The complete protection chain (Quad Z1 + Quad Z2/POTT + 67Q/67N + 87L) eliminates all identified coverage gaps across the TR-29 region map.

References

- [1] J. L. Blackburn and T. J. Domin, *Protective Relaying: Principles and Applications*, 3rd ed. CRC Press, 2006.
- [2] *IEC 60255-121: Measuring Relays and Protection Equipment — Part 121: Functional Requirements for Distance Protection*, International Electrotechnical Commission Std., 2014.